

ALERT meeting

Electron VZ calibration
over the run period

Felix Touchte Codjo

IJCLab

July 6, 2026

- ▶ We use the electron vertex in the AHDC reconstruction

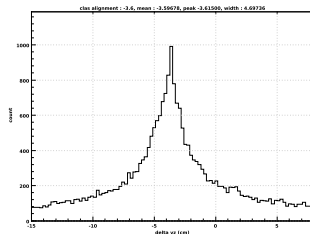
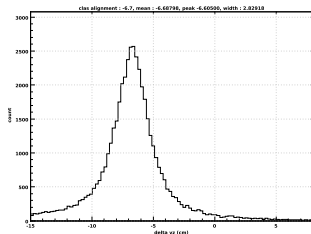
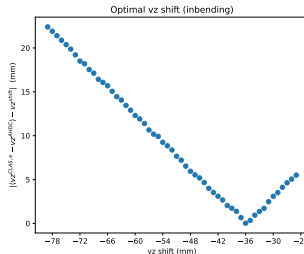
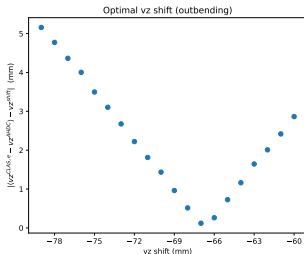
$$(\text{AHDC } v_z) = (\text{CLAS electron } v_z) - (v_z \text{ shift})$$

- ▶ According to the survey ($v_z \text{ shift}$) is -39.94 mm far from the center of CLAS
- ▶ However, looking at the timeline, we know that the v_z reconstruction might be bias depending on the torus configuration
 - up to 30 mm shift between the inbending and outbending configuration
- ▶ We decide to determine ($v_z \text{ shift}$) empirically



Figure: Forward electron v_z (upstream peak value). See [timeline](#)

- ▶ principle : (CLAS electron vz) – (AHDC vz) – (vz shift) should be zero
- ▶ Outbending (left column), Inbending (right column)
- ▶ The optimal values are -67 mm and -36 mm



- Outbending → 2.24 GeV data → elastic cuts : $3.5 < W^2 < 3.8 \text{ GeV}^2$, $\Delta\phi < 20^\circ$, nhits > 6
- Inbending → 10.6 GeV data → nhits > 6, chi2 < 8, no FT electron
- The upstream VZ peak (the one located on the left) vary over the run period. This is mainly due to the change of the torus configuration

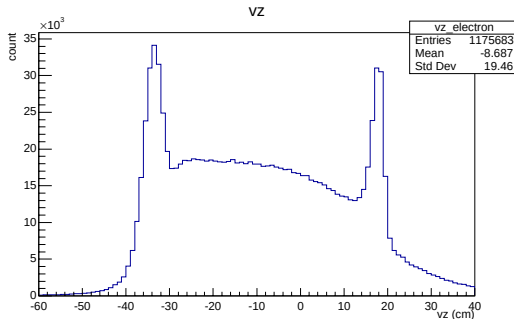


Figure: CLAS electron vertex. Run 22712 (outbending config).